Course notes

Prime numbers

Definition · Prime number. A natural number p is *prime* if it has exactly two positive divisors, 1 and p. Thus 1 is not prime.

Property · Smallest divisor. Every integer $n \geq 2$ has a prime divisor: its smallest divisor greater than or equal to 2. If n is not prime, this divisor p satisfies $p^2 \leq n$.

Proof. Let p be the smallest divisor of n greater than or equal to 2 (it exists, since n is one). If p were not prime, it would have a divisor d with $2 \leq d < p$, which would also divide n: a contradiction. If n is not prime, $n = pq$ with $q \geq 2$ a divisor of n, so $q \geq p$ and $n = pq \geq p^2$. □

Method · Testing whether an integer is prime. To know whether $n \geq 2$ is prime, it is enough to check that no prime p with $p^2 \leq n$ divides n. Example: 211 is divisible neither by 2, 3, 5, 7, 11, nor by 13, and $17^2 = 289 > 211$: 211 is prime.

Property · Infinitude of the primes. There are infinitely many prime numbers.

Proof. Assume there are only finitely many: $p_1, \ldots, p_k$. Let $N = p_1 p_2 \cdots p_k + 1 \geq 2$. By the previous theorem, N has a prime divisor, which is one of the p_i. This p_i divides N and the product $p_1 \cdots p_k$, so it divides their difference 1: this is impossible. The assumption is false. □

Key point · Sieve of Eratosthenes in Python. Cross out the multiples of each prime starting from its square; the numbers not crossed out are prime.

```
def crible(n):
    premier = [True] * (n + 1)
    premier[0] = premier[1] = False
    p = 2
    while p * p <= n:
        if premier[p]:
            for m in range(p * p, n + 1, p):
                premier[m] = False
        p = p + 1
    return [k for k in range(n + 1) if premier[k]]
```

There are 25 prime numbers less than 100.

Property · Prime number and product. Let p be a prime number. If $p \mid ab$, then $p \mid a$ or $p \mid b$. More generally, if p divides a product, it divides one of the factors.

Proof. If $p \nmid a$, then $\gcd(p, a) = 1$, because the only positive divisors of p are 1 and p. Gauss's theorem gives $p \mid b$. The case of a product of several factors follows by induction on their number. □

Prime factorisation

Property · Factorisation. Every integer $n \geq 2$ can be written as a product of primes, $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$, with $p_1 < p_2 < \cdots < p_k$ prime and $\alpha_i \geq 1$. This expression is unique.

Proof. *Existence*, by strong induction. The integer 2 is prime. Assume every integer in $\{2, \ldots, n\}$ can be factorised. If $n+1$ is prime, we are done; otherwise $n+1 = ab$ with $2 \leq a, b \leq n$, and we put the factorisations of a and b together.

Uniqueness. If $p_1 \cdots p_r = q_1 \cdots q_s$ (repeated prime factors), p_1 divides the product of the q_j, hence one of them, which is prime: $p_1 = q_j$. We cancel and start again; the two lists of factors coincide. □

Key point · Factorising in Python. Divide by the smallest divisor $p \geq 2$ as long as possible; it is prime. When $p^2 > n$, the remaining quotient, if it exceeds 1, is prime.

```
def decomposition(n):
    facteurs = []
    p = 2
    while p * p <= n:
        while n % p == 0:
            facteurs.append(p)
            n = n // p
        p = p + 1
    if n > 1:
        facteurs.append(n)
    return facteurs
```

For example, `decomposition(360)` returns `[2, 2, 2, 3, 3, 5]`.

Property · Divisors. If $n = p_1^{\alpha_1} \cdots p_k^{\alpha_k}$, the positive divisors of n are the integers $p_1^{\beta_1} \cdots p_k^{\beta_k}$ with $0 \leq \beta_i \leq \alpha_i$. There are $(\alpha_1+1)(\alpha_2+1)\cdots(\alpha_k+1)$ of them.

Method · GCD by factorisation. The gcd of two integers is the product of the common prime factors, each raised to the smaller of the two exponents. Example: $360 = 2^3 \times 3^2 \times 5$ and $84 = 2^2 \times 3 \times 7$ give $\gcd(360, 84) = 2^2 \times 3 = 12$.

Fermat's little theorem

Property · Fermat's little theorem. Let p be a prime number and a an integer. Then $a^p \equiv a \pmod{p}$. If moreover p does not divide a, then $a^{p-1} \equiv 1 \pmod{p}$.

Proof. *Step 1.* For $1 \leq k \leq p-1$, p divides $\binom{p}{k}$. Indeed $k\binom{p}{k} = p\binom{p-1}{k-1}$, so $p \mid k\binom{p}{k}$; since $1 \leq k < p$, p does not divide k and $\gcd(p, k) = 1$; Gauss gives $p \mid \binom{p}{k}$.

Step 2. By the binomial formula, for every integer a, $(a+1)^p = a^p + \sum_{k=1}^{p-1} \binom{p}{k} a^k + 1 \equiv a^p + 1 \pmod{p}$.

Step 3. Induction on $a \geq 0$: $0^p \equiv 0$, and if $a^p \equiv a$, then $(a+1)^p \equiv a^p + 1 \equiv a + 1 \pmod{p}$. For $a < 0$, choose an integer $b \geq 0$ such that $b \equiv a \pmod{p}$; then $a^p \equiv b^p \equiv b \equiv a$.

Step 4. If $p \nmid a$, then $p \mid a^p - a = a\left(a^{p-1} - 1\right)$ and $\gcd(p, a) = 1$; Gauss gives $p \mid a^{p-1} - 1$. □

Example 1 · A remainder. Remainder of 3^{2026} in the division by 11: 11 is prime and does not divide 3, so $3^{10} \equiv 1 \pmod{11}$. Since $2{,}026 = 10 \times 202 + 6$, $3^{2026} \equiv 3^6 = 729 \equiv 3 \pmod{11}$ (because $729 = 66 \times 11 + 3$).

Watch out. The converse is false: $2^{340} \equiv 1 \pmod{341}$, whereas $341 = 11 \times 31$.